



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Aung Myat
Sept 26, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- **Methodology**

- Collected Falcon 9 launch data using the **SpaceX REST API** and **web scraping**
- Cleaned and transformed the data for **EDA, SQL analysis, and visualization**
- Developed interactive analytics using **Folium** and **Plotly Dash**
- Built and tuned **Logistic Regression, SVM, Decision Tree, and KNN** classifiers using 10-fold cross-validation

- **Key Results**

- Landing success generally improved with flight experience and over time
- Launch site, orbit type, and payload mass were associated with landing performance
- Logistic Regression, SVM, and KNN each achieved 83.33% test accuracy
- SVM was selected for final reporting, with 84.82% cross-validation accuracy
- The selected model identified all successful landings in the test set, with false positives as the main error source

Introduction

- Project background and context
 - SpaceX reduces Falcon 9 launch costs by recovering and reusing the first-stage booster. Therefore, the ability to determine whether the first stage will land successfully is important for understanding launch economics and evaluating the competitiveness of a launch mission.
- Problems you want to find answers
 - What factors—such as **launch site, payload mass, orbit, booster version, and flight experience**—are associated with successful landings?
 - How does **landing success vary across launch sites, orbit types, and over time**?
 - Can historical launch data be used to **predict whether a Falcon 9 first stage will land successfully**?
 - What insights can support an alternative launch provider in evaluating **mission cost and competitiveness**?

Section 1

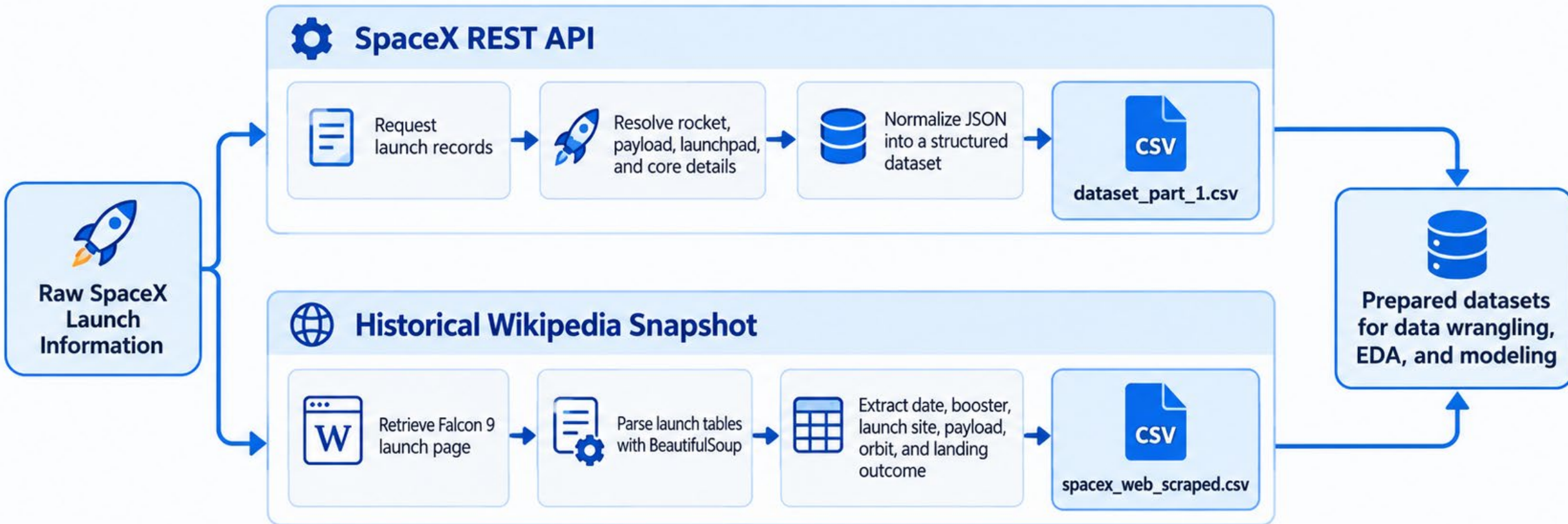
Methodology

Methodology

- Data collection methodology:
 - SpaceX REST API
 - Web Scrapping of Historical Launch Data
- Perform data wrangling
 - Cleaned and transformed launch records and Created binary landing-success target
- Perform exploratory data analysis (EDA)
 - Python Visualization and SQL-based analysis
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Classification model development
 - Hyperparameter tuning and model evaluation

Data Collection

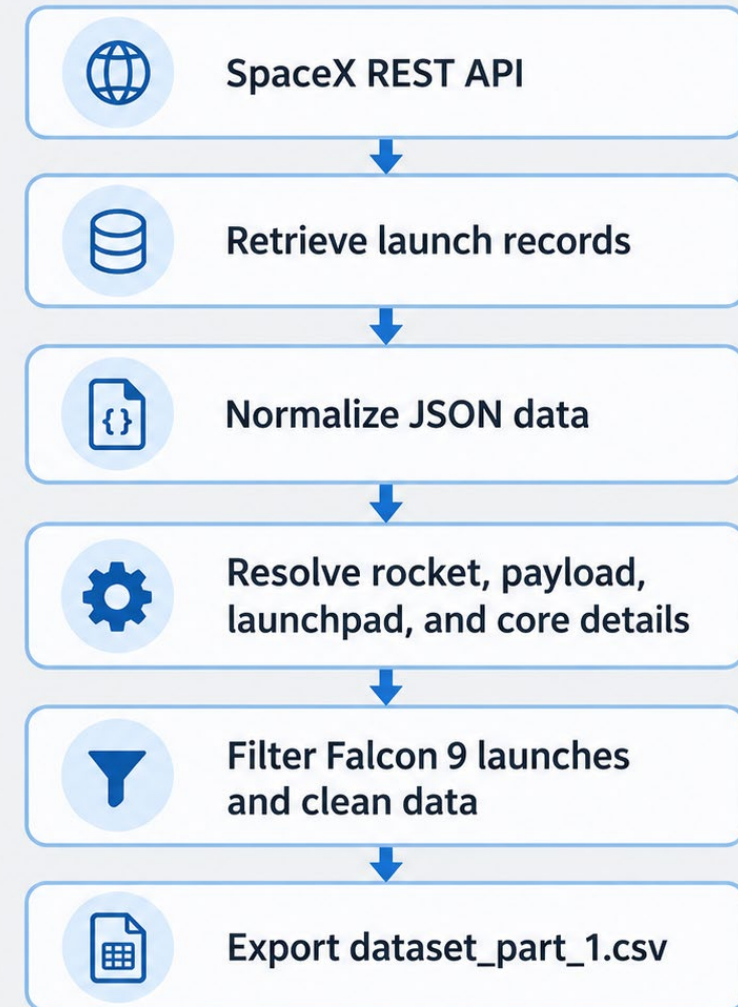
Data Collection Process



Data Collection – SpaceX API

- **SpaceX REST API workflow**
 - Retrieved Falcon launch records using the **SpaceX REST API**
 - Sent HTTP GET requests with Python **requests**
 - Parsed and normalized JSON responses into a **Pandas DataFrame**
 - Resolved **rocket, payload, launchpad, and core** information through related API endpoints
 - Filtered the dataset to retain **Falcon 9 launches**
 - Cleaned and structured the data for subsequent analysis
 - Exported the processed dataset as **dataset_part_1.csv**
- **Completed notebook:**
 - https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/01_SpaceX_Data_Collection_API_completed.ipynb

SpaceX API Data Collection Flow



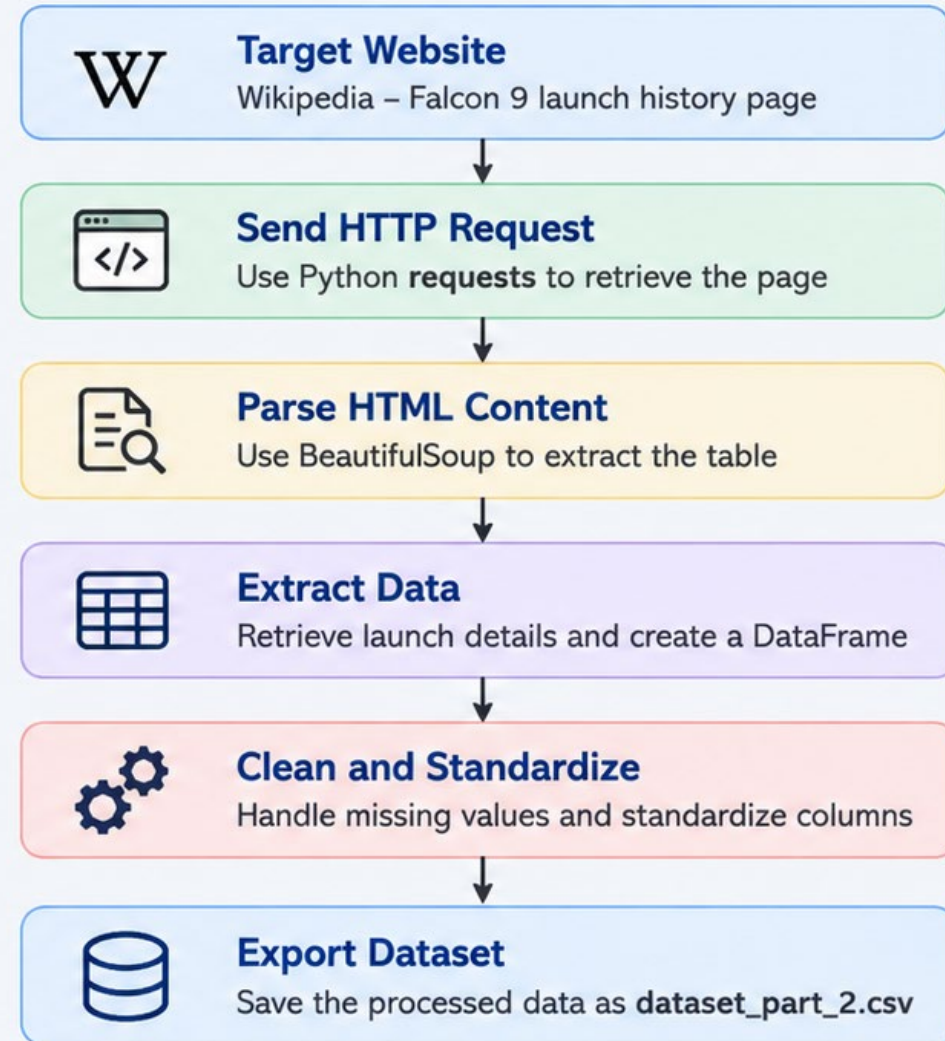
Data Collection - Scraping

- **Web Scraping Workflow**

- Retrieved a fixed **Falcon 9 launch-history Wikipedia snapshot** using Python requests
- Parsed the HTML launch tables using **BeautifulSoup**
- Extracted launch-level fields, including **date, booster, launch site, payload, orbit, and landing outcome**
- Cleaned annotations and inconsistent formatting
- Converted the extracted records into a **Pandas DataFrame** and exported the structured dataset

- **Completed notebook:**

- https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/O2_SpaceX_Web_Scraping_completed.ipynb



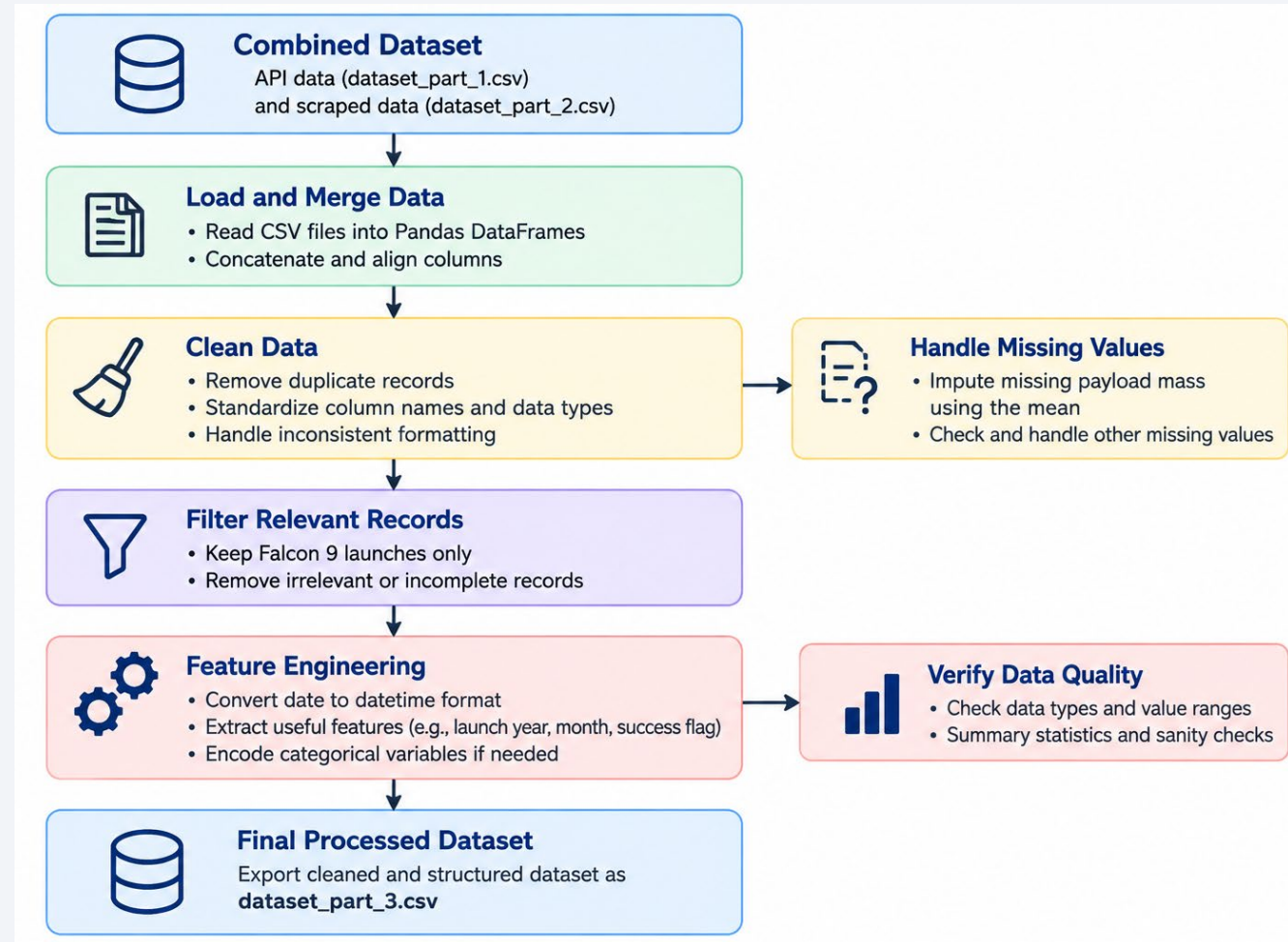
Data Wrangling

- Data Wrangling Process

- Loaded the collected **Falcon 9 launch dataset** into Pandas
- Reviewed **landing outcomes, launch-site frequencies, and orbit frequencies**
- Converted landing outcomes into a binary target variable, **Class**
 - 1 = successful landing
 - 0 = unsuccessful landing
- Verified the distribution of the processed target variable
- Exported the wrangled dataset as **dataset_part_2.csv**
- Prepared the dataset for **EDA and machine-learning analysis**

- Completed Notebook

- https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/O3_SpaceX_Data_Wrangling_complete.d.ipynb



EDA with Data Visualization

- **Charts and analytical purpose**

- **Scatter plots – Flight Number vs. Launch Site**
Examined how landing success changed with **flight experience across launch sites**.
- **Scatter plots – Payload Mass vs. Launch Site**
Evaluated whether **payload mass and launch location** were associated with landing success.
- **Bar chart – Success Rate by Orbit Type**
Compared the **landing success rate across different orbit categories**.
- **Scatter plots – Flight Number vs. Orbit Type**
Investigated relationships among **orbit type, flight experience, and landing outcome**.
- **Scatter plots – Payload Mass vs. Orbit Type**
Assessed whether certain **payload ranges and orbit types** were associated with successful landings.
- **Line chart – Yearly Launch Success Trend**
Visualized how the **Falcon 9 landing success rate evolved over time**.
- **Visualization tools:** Pandas, Matplotlib, Seaborn

- **Completed Notebook**

- https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/04_SpaceX_EDA_Data_Visualization_completed.ipynb

EDA with SQL

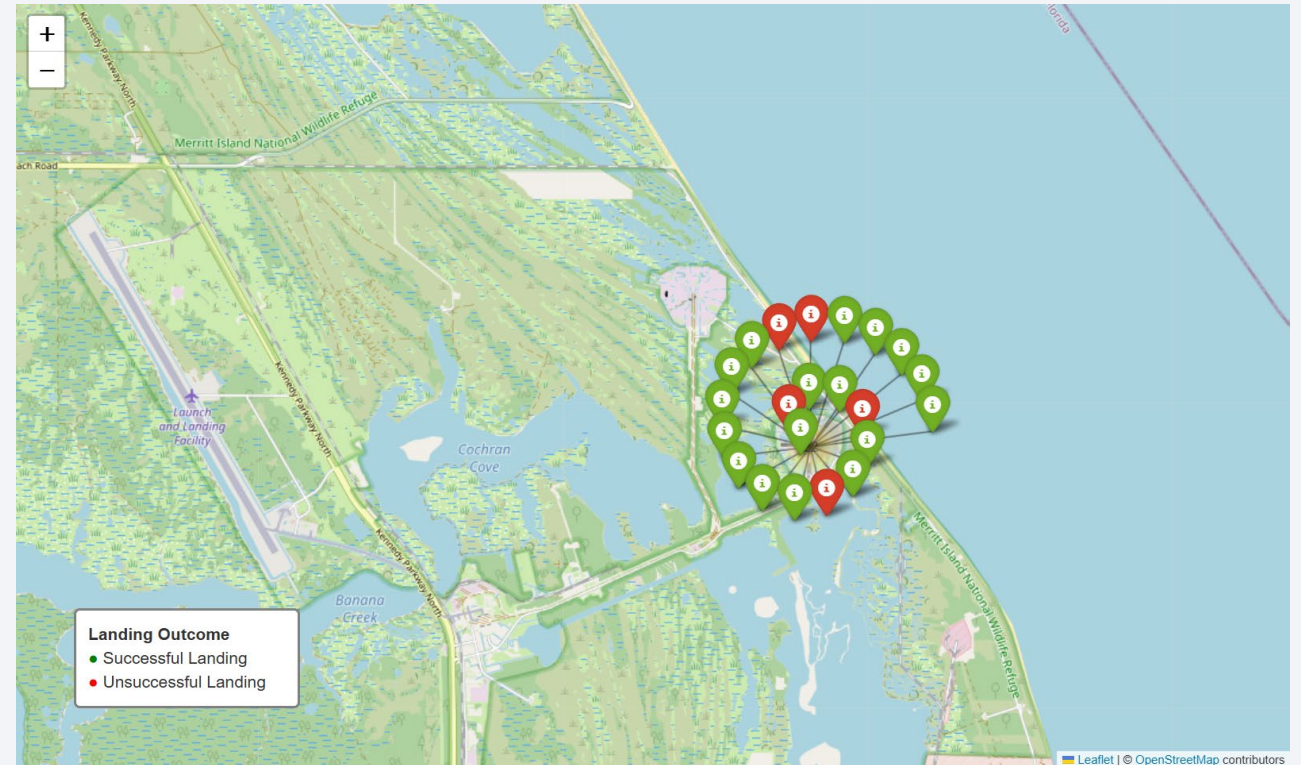
- the SQL queries performed
 - Identified all **unique launch sites**
 - Retrieved launch records from **KSC launch sites**
 - Calculated the **total payload mass for NASA (CRS)**
 - Calculated the **average payload mass for F9 v1.1**
 - Identified the **first successful drone-ship landing date**
- Completed Notebook
 - https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/05_SpaceX_EDA_SQL_completed.ipynb
- the SQL queries performed
 - Identified booster versions with successful landings for **4,000–6,000 kg payloads**
 - Counted launches by **mission outcome**
 - Identified the booster version carrying the **maximum payload**
 - Retrieved **2017 successful ground-pad landing records**
 - Ranked **landing outcomes by frequency** for the specified date range

Build an Interactive Map with Folium

- Map Objects Added
 - Circle markers and labels for Falcon 9 launch-site locations
 - Marker clusters for mission-level launch records
 - Color-coded markers
 - Green = successful landing
 - Red = unsuccessful landing
 - **Mouse-position tool** – inspect latitude and longitude interactively
 - **Feature markers and distance lines** – show nearby coastline, railway, highway, and city
 - Distance lines connecting the selected launch site to:
 - Coastline
 - Railway
 - Highway
 - Nearby city

Purpose

- Visualize the **geographic distribution** of launch facilities
- Compare **landing outcomes by location**
- Reduce marker overlap while retaining individual mission records
- Examine the **proximity of launch sites to geographic and transportation features**



Completed notebook:

https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/07_SpaceX_Interactive_Map_Folium.ipynb

Build a Dashboard with Plotly Dash

Interactive Dashboard Components

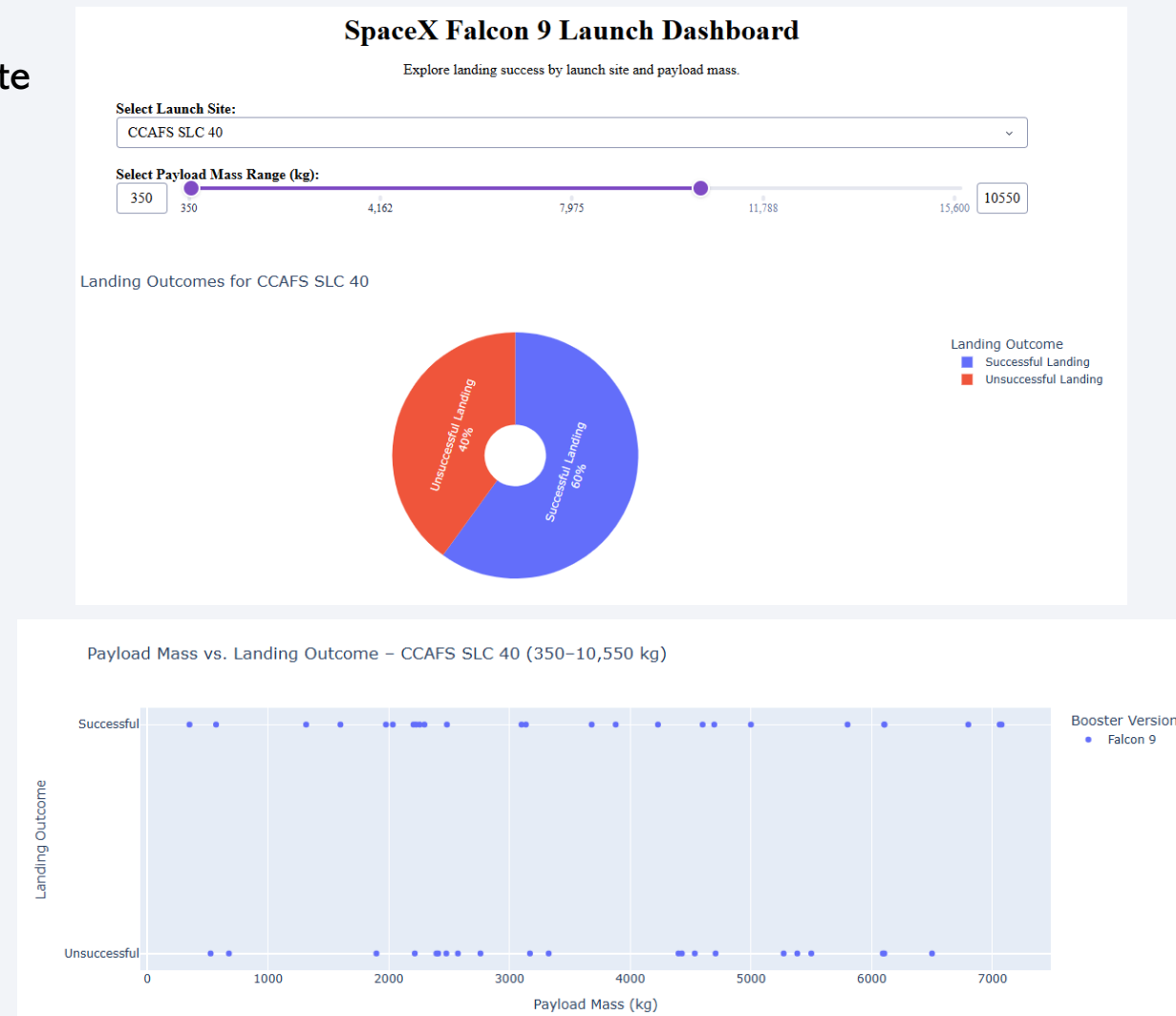
- **Launch-site dropdown** – select all sites or one launch site
- **Payload-mass range slider** – filter launches by payload
- **Landing-outcome pie chart** – compare successful vs. unsuccessful landings
- **Payload vs. landing-outcome scatter plot** – examine payload effects
- **Booster-version grouping** – compare Falcon 9 configurations
- **Dash callbacks** – update charts automatically

Purpose

- Compare landing performance across **launch sites**
- Explore **payload mass–landing outcome** relationships
- Examine differences among **booster versions**
- Enable interactive filtering beyond static EDA and SQL

Completed notebook:

[https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/08 SpaceX Interactive Dashboard Plotly Dash.ipynb](https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/08%20SpaceX%20Interactive%20Dashboard%20Plotly%20Dash.ipynb)



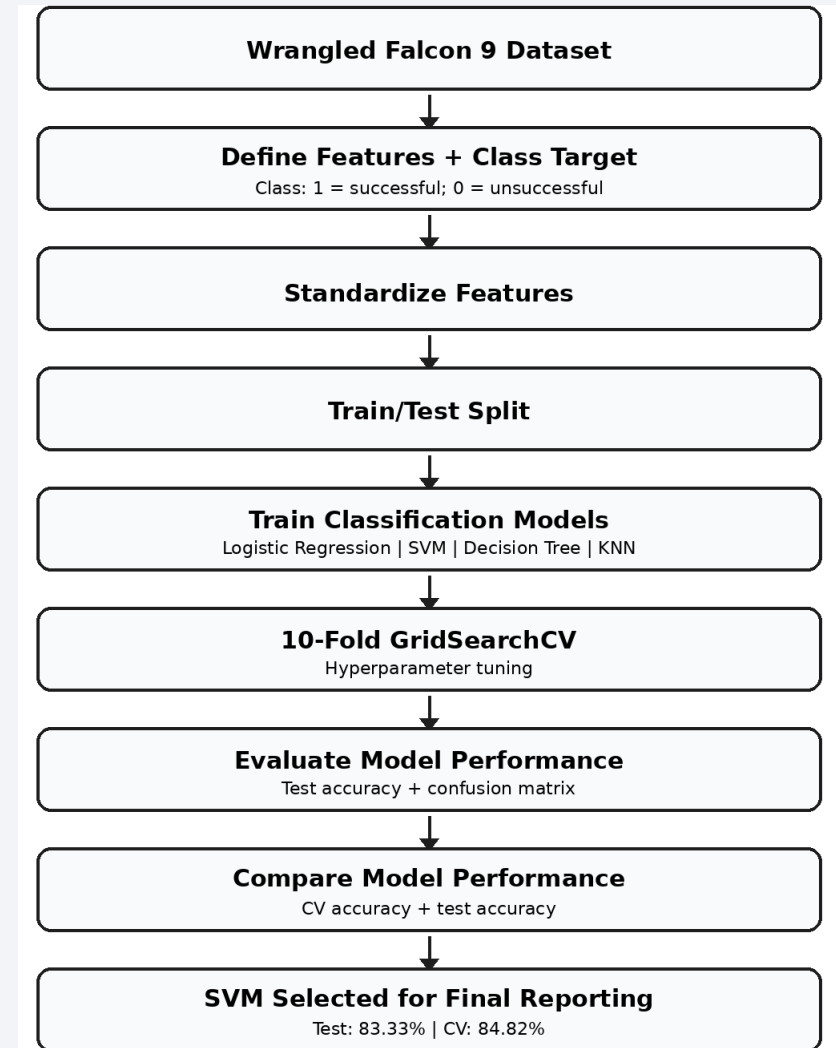
Predictive Analysis (Classification)

Model Development & Evaluation

- **Target:** Class — 1 = successful landing; 0 = unsuccessful landing
- Standardized predictor variables and performed a **train/test split**
- Evaluated four classifiers: **Logistic Regression, SVM, Decision Tree, and KNN**
- Tuned hyperparameters using **GridSearchCV** with **10-fold cross-validation**
- Assessed performance using **CV accuracy, test accuracy, and confusion matrices**
- Logistic Regression, SVM, and KNN achieved **83.33% test accuracy**
- **SVM selected for final reporting** based on the highest CV accuracy among the tied test leaders (**84.82%**)

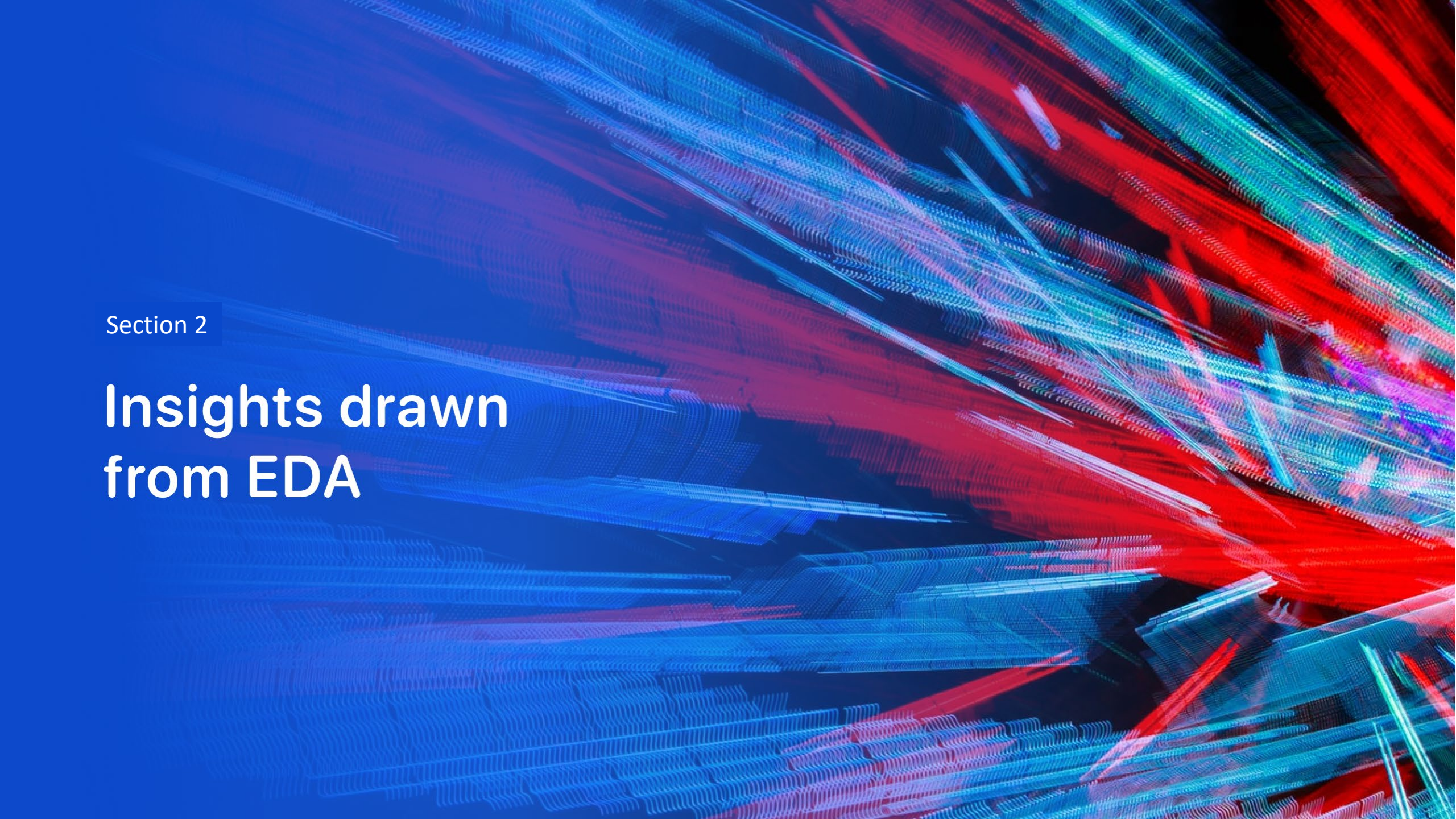
Completed notebook:

https://github.com/Zephyr923/SpaceX-Falcon9-Capstone/blob/main/06_SpaceX_Machine_Learning_completed.ipynb



Results

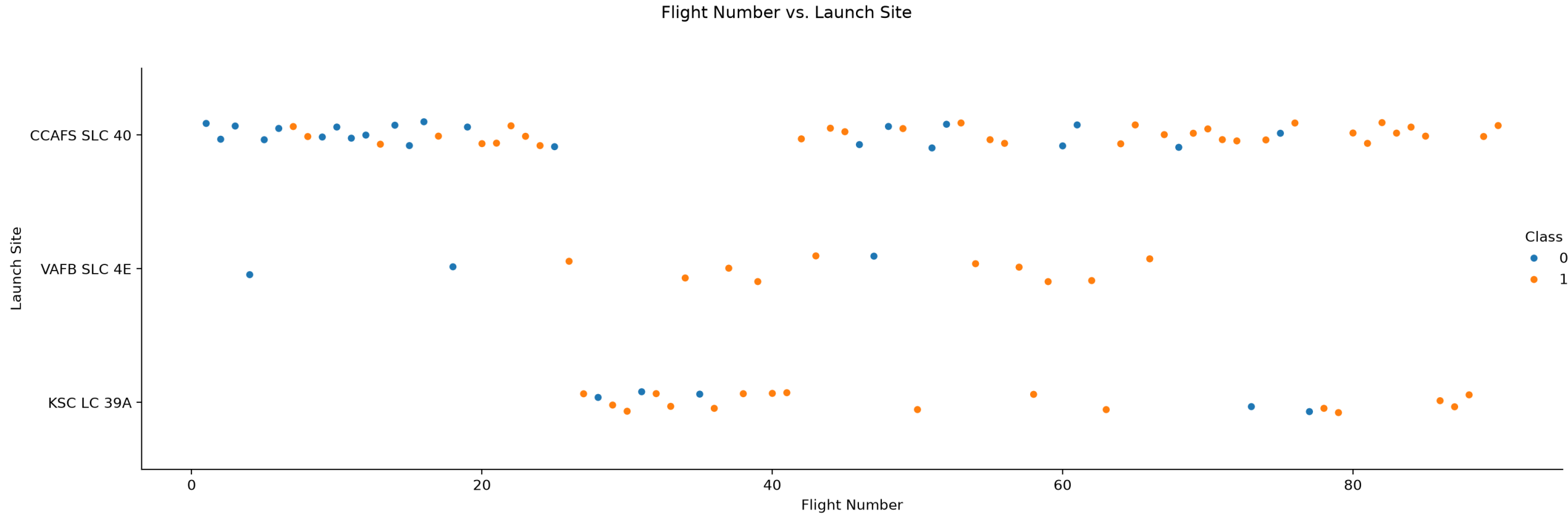
- Exploratory data analysis results
 - Visualization and SQL findings
- Interactive analytics demo in screenshots
 - Folium maps and Plotly Dash dashboard
- Predictive analysis results
 - Classification model comparison and confusion matrices
- Key Findings and Insights
 - Overall patterns relevant to Falcon 9 landing success

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

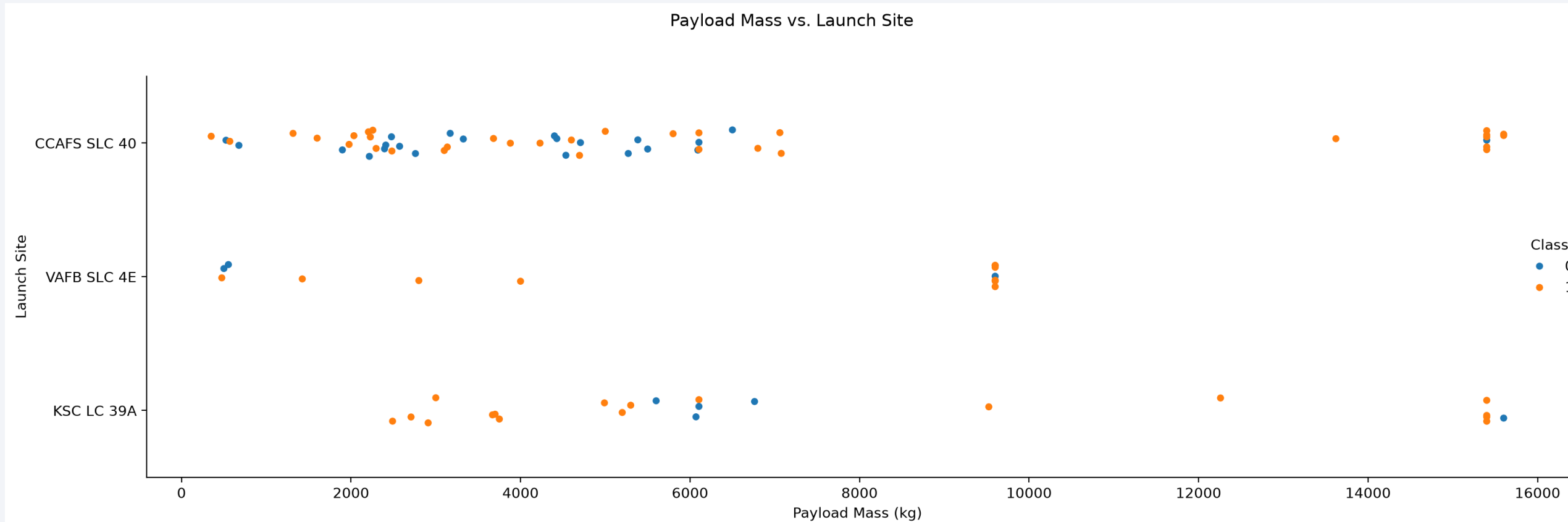


Falcon 9 landing outcomes by flight number and launch site; Successful and unsuccessful landings are distinguished by the Class variable.

- **Key Findings**

- Higher flight numbers show **more successful landings**
- Earlier flights show **more failures**
- Performance differs across **launch sites**

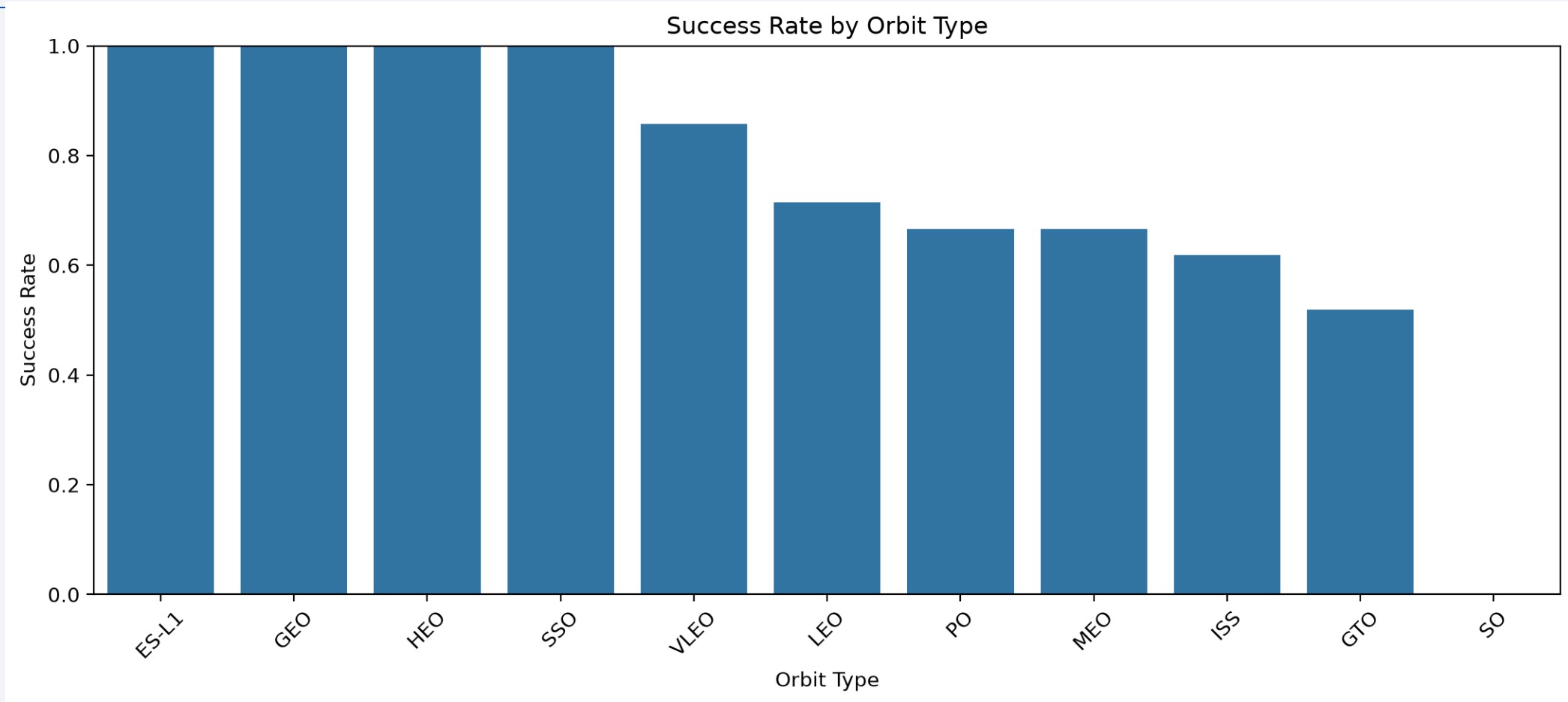
Payload vs. Launch Site



- **Key Findings**

- Landing success varies with **payload mass**
- **KSC LC-39A** shows strong success across a broad payload range
- Some launch sites show **more failures at lower payloads**

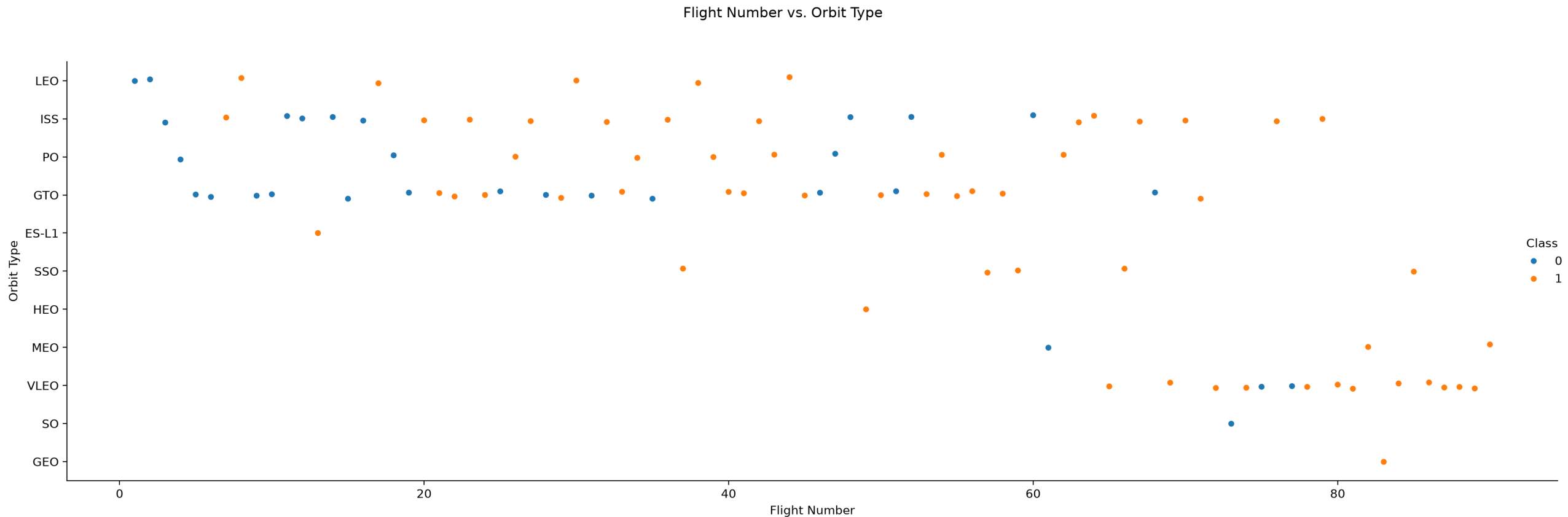
Success Rate vs. Orbit Type



Key Findings

- Landing success varies considerably by **orbit**
- Several orbit types show **high success rates**
- Orbit type is an important factor in **landing performance**

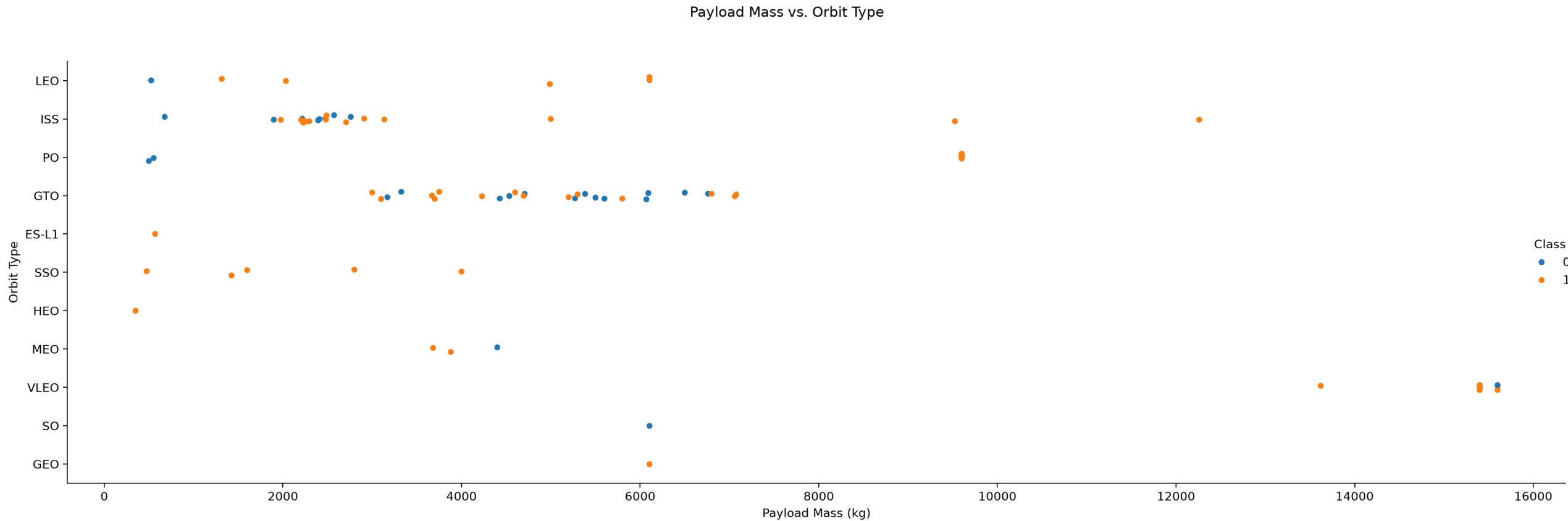
Flight Number vs. Orbit Type



Key Findings

- Later flights show **more successful landings**
- Success improves across several **orbit categories**
- Flight experience appears related to **landing reliability**

Payload vs. Orbit Type



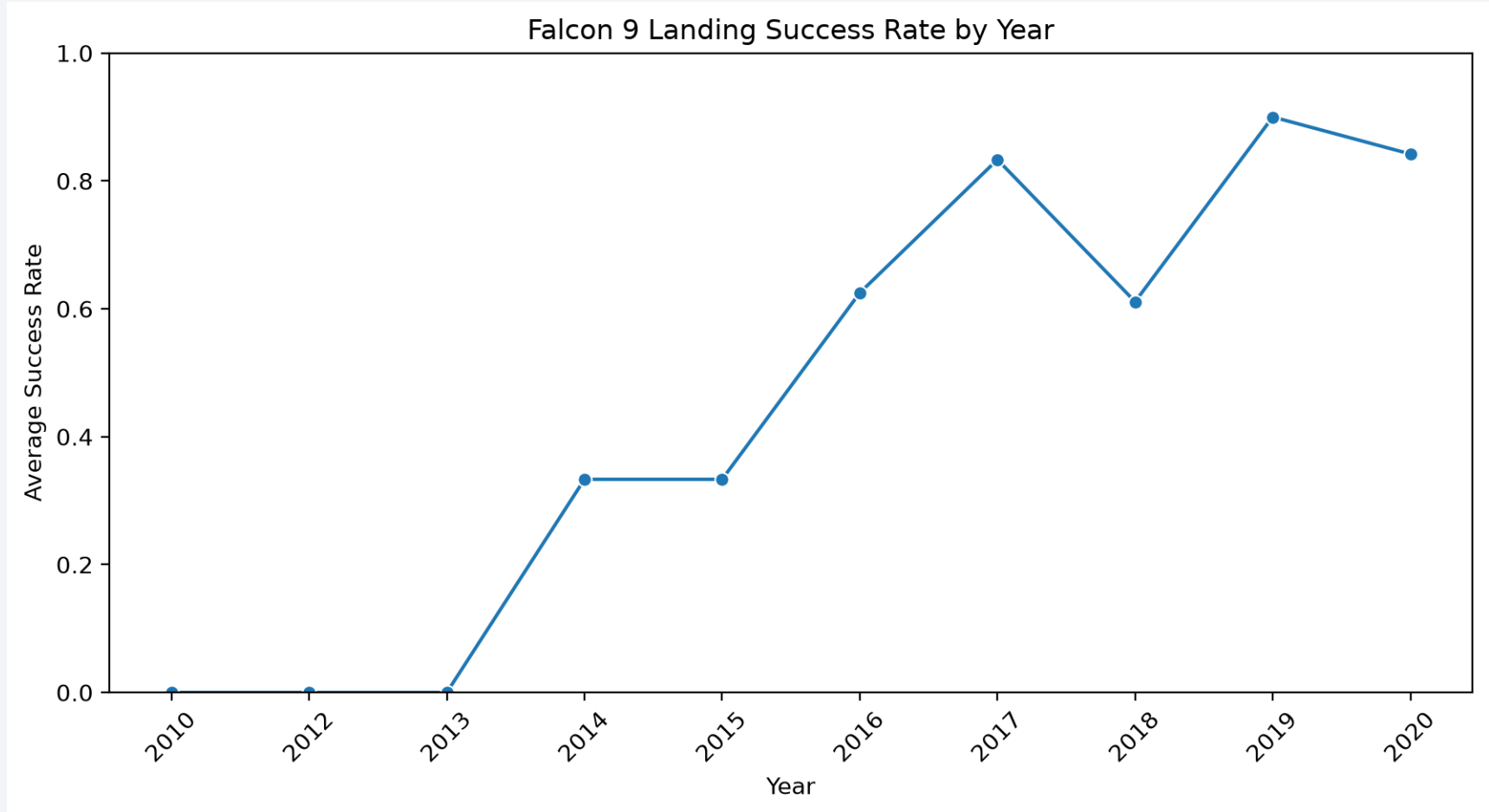
Key Findings

- Payload ranges differ substantially across **orbit types**
- Landing success depends on both **payload and orbit**
- No single payload range guarantees **landing success**

Launch Success Yearly Trend

Key Findings

- Landing success generally **increased over time**
- Earlier years show **lower success rates**
- Later missions demonstrate improved **landing reliability**



All Launch Site Names

- **Query:** Identify all unique launch sites in the dataset.
- **Key Result:** Four distinct launch-site names were identified in the SQL dataset.

```
[21]: query = """  
      SELECT DISTINCT Launch_Site  
      FROM SPACEXTABLE;  
      """  
  
      task1 = pd.read_sql_query(query, con)  
      task1
```

```
[21]:
```

	Launch_Site
0	CCAFS LC-40
1	VAFB SLC-4E
2	KSC LC-39A
3	CCAFS SLC-40

Launch Site Names Begin with 'KSC'

Display 5 records where launch sites begin with the string 'KSC'

```
[24]: ##sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'KSC%' LIMIT 5;  
query = ""  
SELECT *  
FROM SPACEXTABLE  
WHERE Launch_Site LIKE 'KSC%'  
LIMIT 5;  
""  
  
task2 = pd.read_sql_query(query, con)  
task2
```

```
[24]:
```

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
1	2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar	Success	No attempt
2	2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES	Success	Success (drone ship)
3	2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	Success	Success (ground pad)
4	2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat	Success	No attempt

- **Query:** Retrieve launch records where the launch-site name begins with **KSC**.
- **Key Result:** Five KSC launch records were returned for review.

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[25]: ##sql SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
query = """  
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass  
FROM SPACEXTABLE  
WHERE Customer = 'NASA (CRS)';  
"""
```

```
task3 = pd.read_sql_query(query, con)  
task3
```

```
[25]:
```

	Total_Payload_Mass
0	45596

- **Query:** Calculate the total payload mass carried for **NASA (CRS)** missions.
- **Key Result:** 45596 kg

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
[26]: %%sql SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
query = ""
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload_Mass
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
""

task4 = pd.read_sql_query(query, con)
task4
```

```
[26]:
```

	Average_Payload_Mass
0	2928.4

- **Query:** Calculate the average payload mass carried by **Falcon 9 v1.1**.
- **Key Result:** Average payload mass = **2928.4 kg**

First Successful Drone-Ship Landing Date

```
[27]: %%sql SELECT MIN(Date) AS First_Successful_Drone_Ship_Landing FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (drone ship)';
query = """
SELECT MIN(Date) AS First_Successful_Drone_Ship_Landing
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)';
"""

task5 = pd.read_sql_query(query, con)
task5
```

```
[27]: First_Successful_Drone_Ship_Landing
0                2016-04-08
```

- **Query:** Identify the earliest successful landing on a **drone ship**.
- **Key Result: First successful drone-ship landing: 2016-04-08**

Successful Ground-Pad with Payload between 4000 and 6000

```
[28]: %%sql SELECT Booster_Version FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)' AND PAYLOAD_MASS_KG_ > 4000 AND PAYLOAD_MASS_KG_ < 6000
query = """
SELECT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)'
AND PAYLOAD_MASS_KG_ > 4000
AND PAYLOAD_MASS_KG_ < 6000;
"""

task6 = pd.read_sql_query(query, con)
task6
```

```
[28]:
```

	Booster_Version
0	F9 FT B1032.1
1	F9 B4 B1040.1
2	F9 B4 B1043.1

- **Query:** Identify booster versions that successfully landed on a **ground pad** with payload mass between **4,000 and 6,000 kg**.
- **Key Result:** Booster versions: F9 FT B1032.1, F9 B4 B1040.1, F9 B4 B1043.1

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
[30]: %%sql SELECT Mission_Outcome, COUNT(*) AS Outcome_Count FROM SPACEXTABLE GROUP BY Mission_Outcome;
query = """
SELECT Mission_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
"""

task7 = pd.read_sql_query(query, con)
task7
```

```
[30]:
```

	Mission_Outcome	Outcome_Count
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

- **Query:** Count the number of missions for each **mission outcome** category.
- **Key Result:**
Mission outcomes were grouped and counted to compare successful and unsuccessful missions.

Boosters Carried Maximum Payload

- **Query:** Identify the booster version(s) associated with the **maximum payload mass** in the dataset.

```
[31]: query = """
      SELECT Booster_Version
      FROM SPACEXTABLE
      WHERE PAYLOAD_MASS_KG_ = (
            SELECT MAX(PAYLOAD_MASS_KG_)
            FROM SPACEXTABLE
      );
      """

task8 = pd.read_sql_query(query, con)
task8
```

```
[31]:
```

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2017 Launch Records

- **Query:** Retrieve all **successful ground-pad landings in 2017** with month, booster version, and launch site.
- **Key Result:**
The query identifies the 2017 missions that successfully landed on a ground pad and shows when and where they occurred.

```
[32]: query = """
      SELECT
          substr(Date, 6, 2) AS Month,
          Landing_Outcome,
          Booster_Version,
          Launch_Site
      FROM SPACEXTABLE
      WHERE substr(Date, 1, 4) = '2017'
          AND Landing_Outcome = 'Success (ground pad)';
      """

      task9 = pd.read_sql_query(query, con)
      task9
```

```
[32]:
```

	Month	Landing_Outcome	Booster_Version	Launch_Site
0	02	Success (ground pad)	F9 FT B1031.1	KSC LC-39A
1	05	Success (ground pad)	F9 FT B1032.1	KSC LC-39A
2	06	Success (ground pad)	F9 FT B1035.1	KSC LC-39A
3	08	Success (ground pad)	F9 B4 B1039.1	KSC LC-39A
4	09	Success (ground pad)	F9 B4 B1040.1	KSC LC-39A
5	12	Success (ground pad)	F9 FT B1035.2	CCAFS SLC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **Query:** Rank landing outcomes by frequency between **2010-06-04** and **2017-03-20**.
- **Key Result:**
Landing outcomes were ranked from most to least frequent over the specified period.

```
[33]: query = """
SELECT
    Landing_Outcome,
    COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
"""

task10 = pd.read_sql_query(query, con)
task10
```

```
[33]:
```

	Landing_Outcome	Outcome_Count
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of white clouds and a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The horizon line is visible, separating the dark sky from the Earth's surface.

Section 3

Launch Sites Proximities Analysis

Geographic distribution of Falcon 9 launch sites

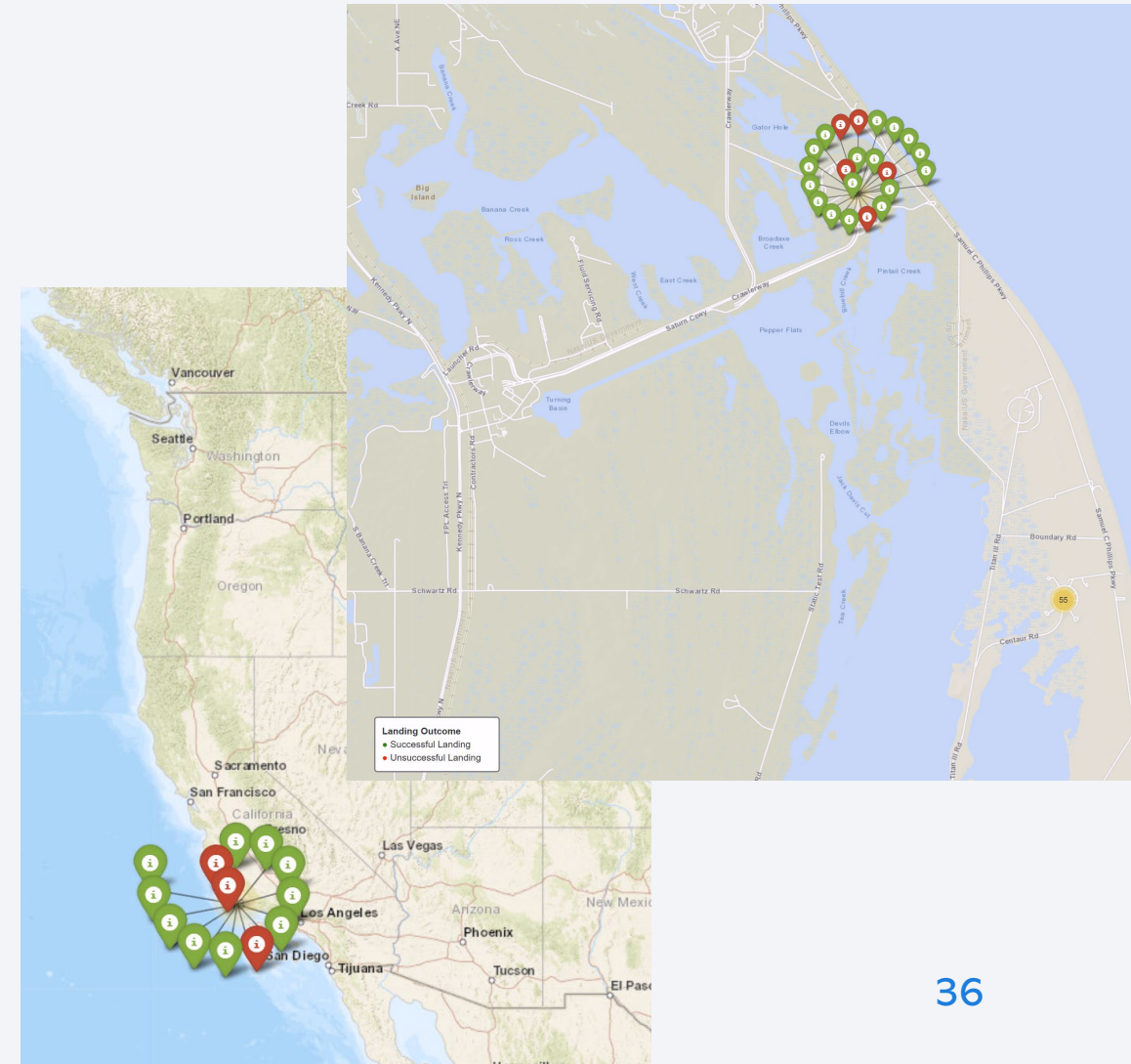
- Geographic distribution of Falcon 9 launch sites
- Key Findings
 - Falcon 9 launches occurred from sites in **Florida and California**
 - Multiple launch facilities are concentrated on the **Florida coast**
 - Mapping establishes the geographic basis for later landing analysis



Geographic distribution of Falcon 9 launch sites

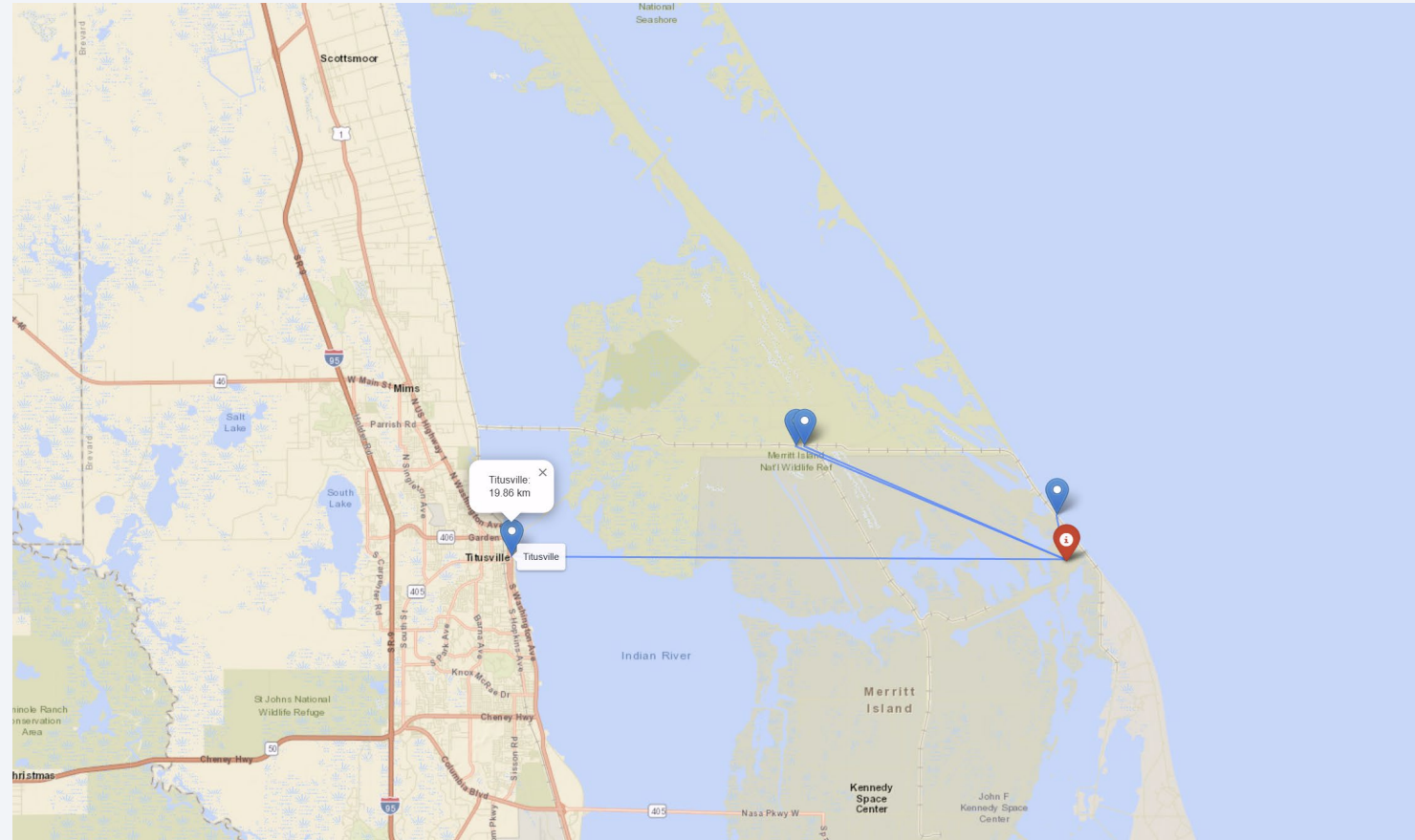
Successful and unsuccessful Falcon 9 landing outcomes by launch site

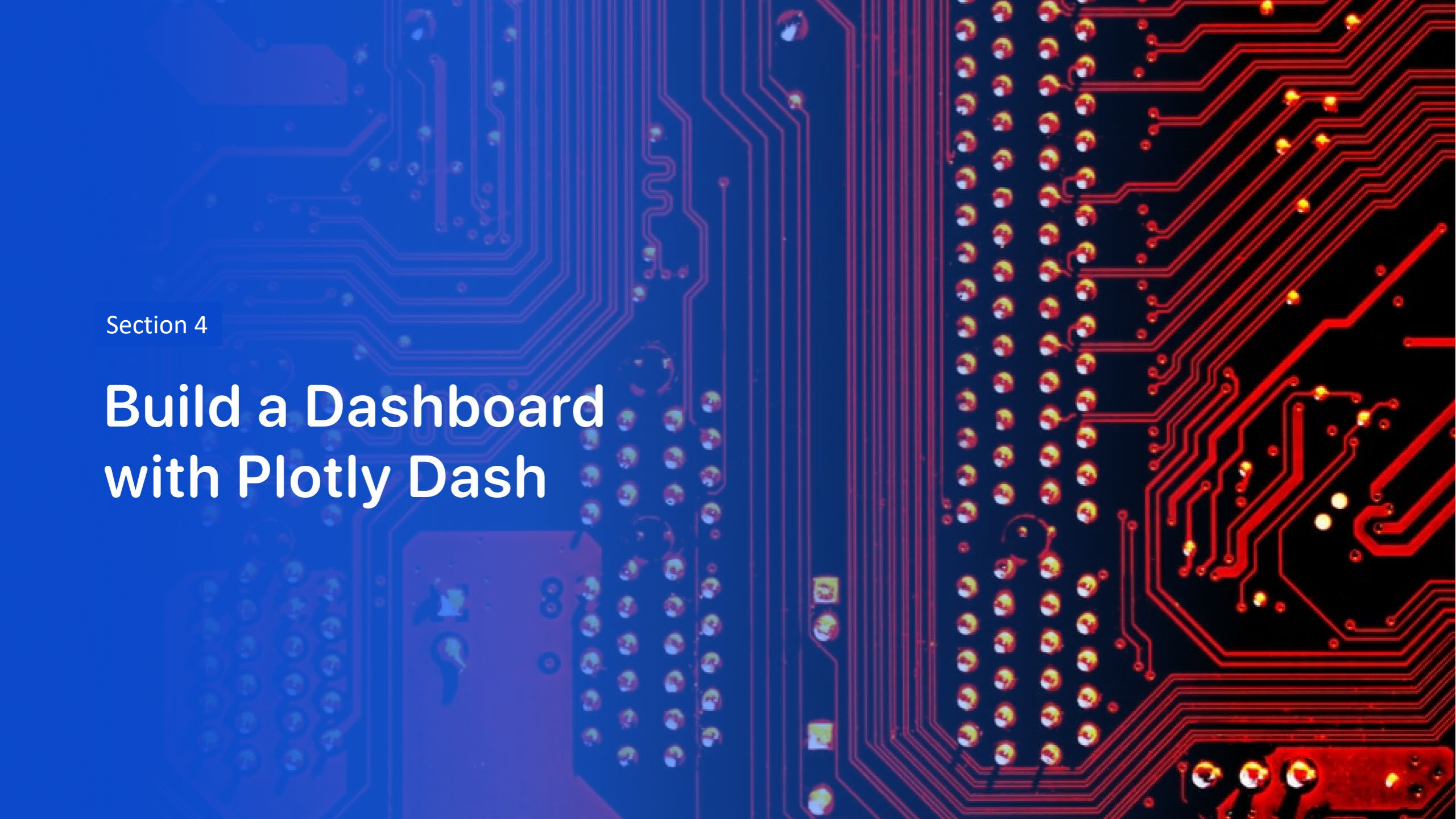
- Successful and unsuccessful Falcon 9 landing outcomes by launch site
- **Key Findings**
 - **Green** markers indicate successful landings
 - **Red** markers indicate unsuccessful landings
 - Landing performance can be explored by **launch location and mission**



Proximity of the selected launch site to nearby geographic and transportation features

- Proximity of the selected launch site to nearby geographic and transportation features
- **Key Findings**
 - The selected launch site is located close to the **coastline**
 - Nearby **railway, highway, and city** connections were identified
 - Distance lines quantify the site's proximity to surrounding features





Section 4

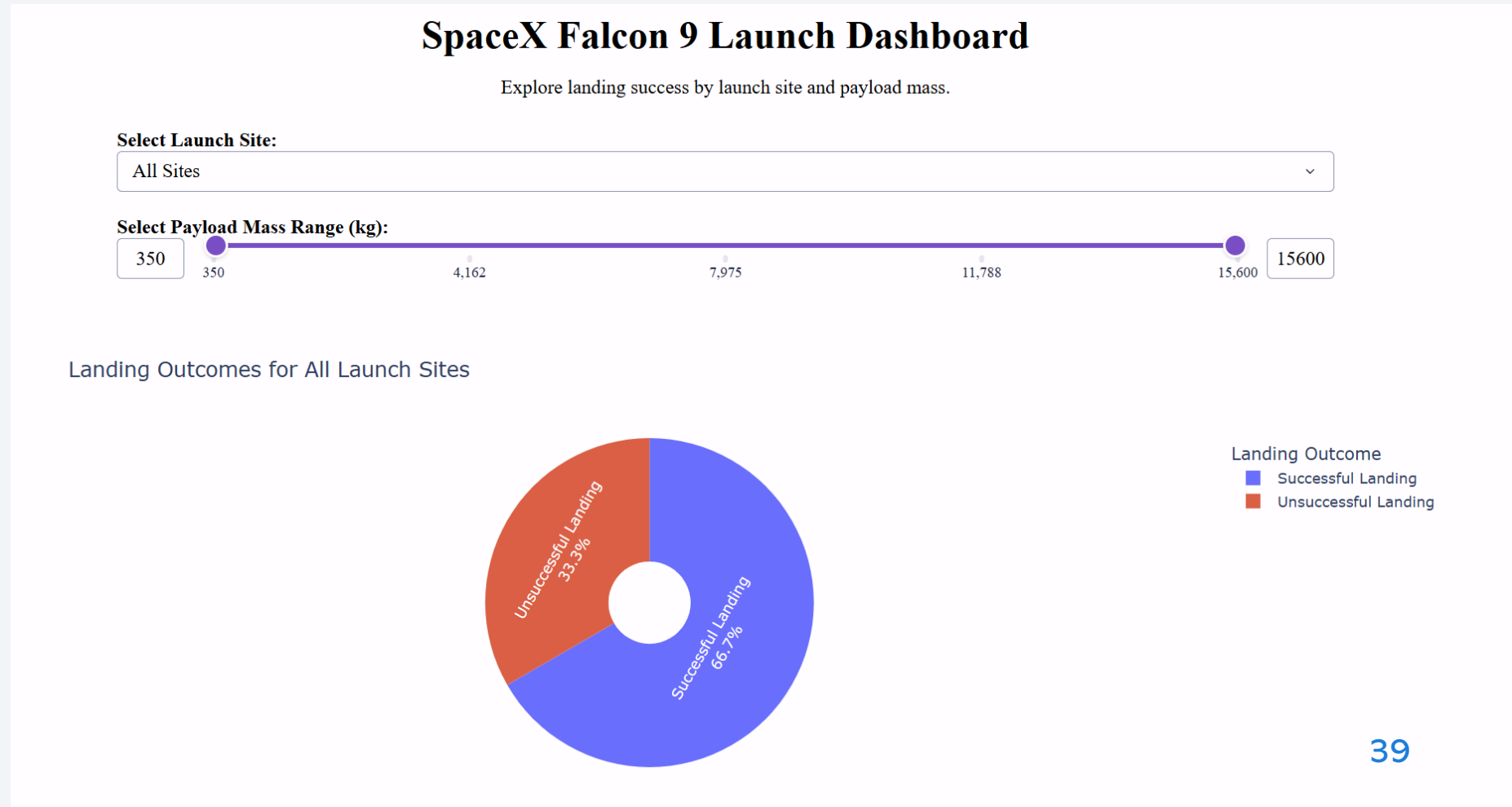
Build a Dashboard with Plotly Dash

Falcon 9 Landing Success – All Launch Sites

Falcon 9 Landing Success – All Launch Sites

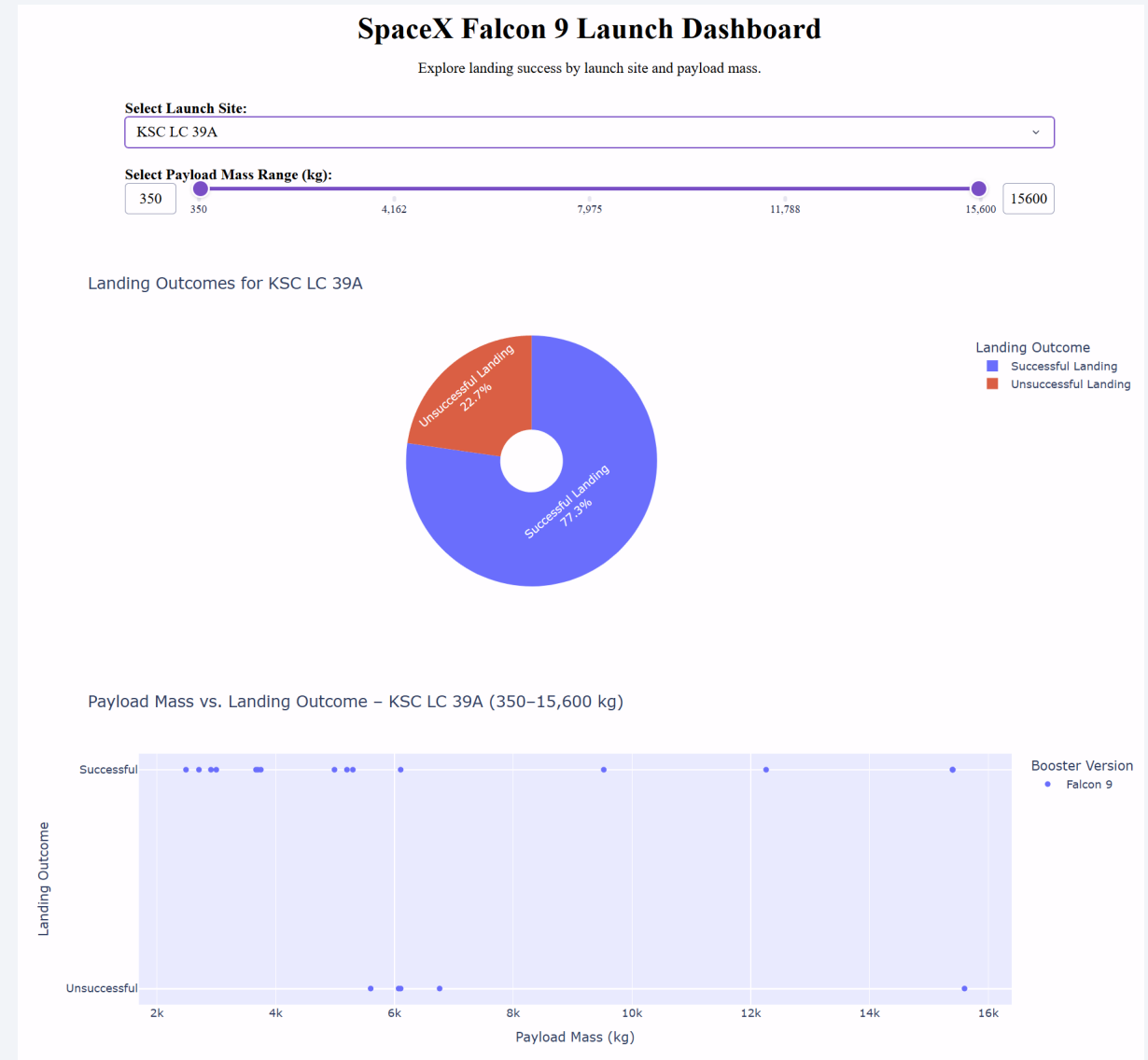
Key Findings

- **66.7%** of launches resulted in successful landings
- Successful landings were about **twice as frequent** as failures
- Overall performance provides the baseline for site-level comparison



Highest Landing Success Rate by Launch Site

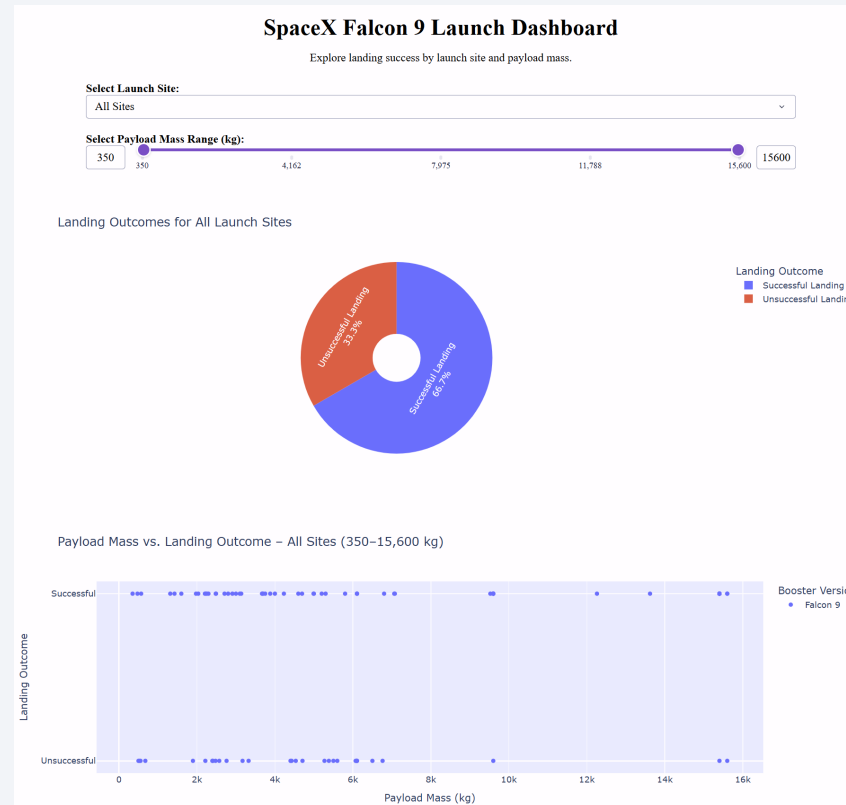
- KSC LC-39A – Landing Success Distribution
- **Key Findings**
- **KSC LC 39A** achieved the highest landing success ratio
- Successful landings accounted for **77.3 %** of launches at this site
- Site-level performance exceeds the **66.7% overall success rate**



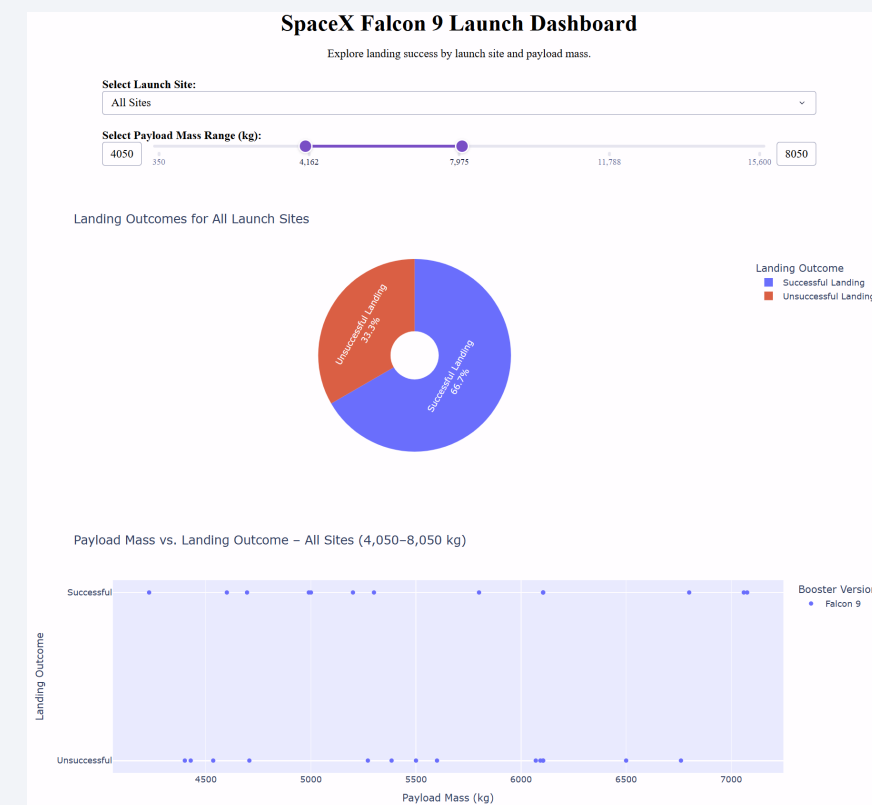
Payload Mass vs. Landing Outcome

- Interactive relationship between payload mass, landing outcome, and booster version
- **Key Findings**
 - Landing outcomes vary across the **payload-mass range**
 - Successful and unsuccessful missions occur at overlapping payload masses
 - Interactive filtering reveals performance within **specific payload ranges and booster versions**

Full payload range



Selected payload range



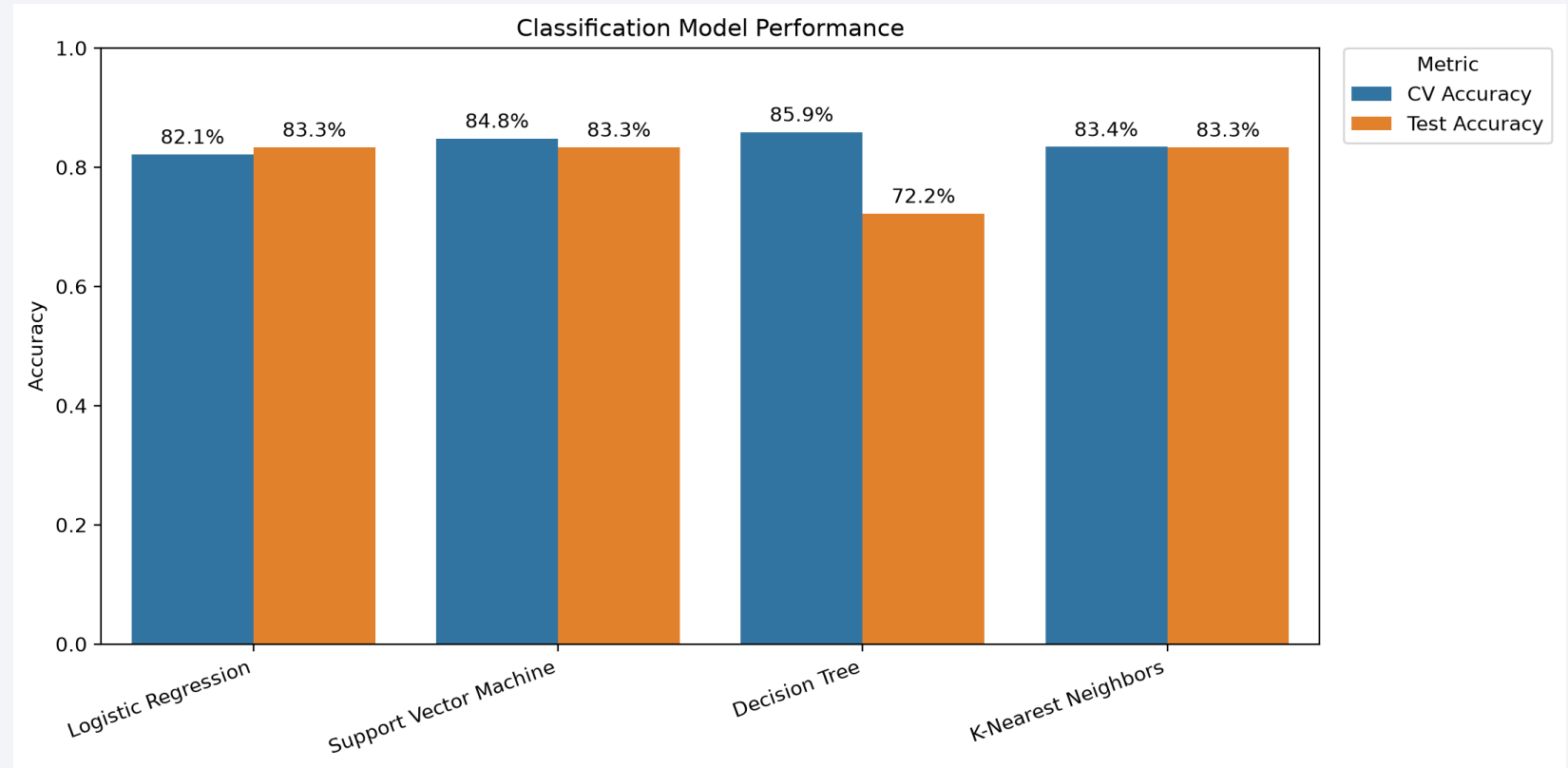
Section 5

Predictive Analysis (Classification)

Classification Model Performance

- **Key Findings**

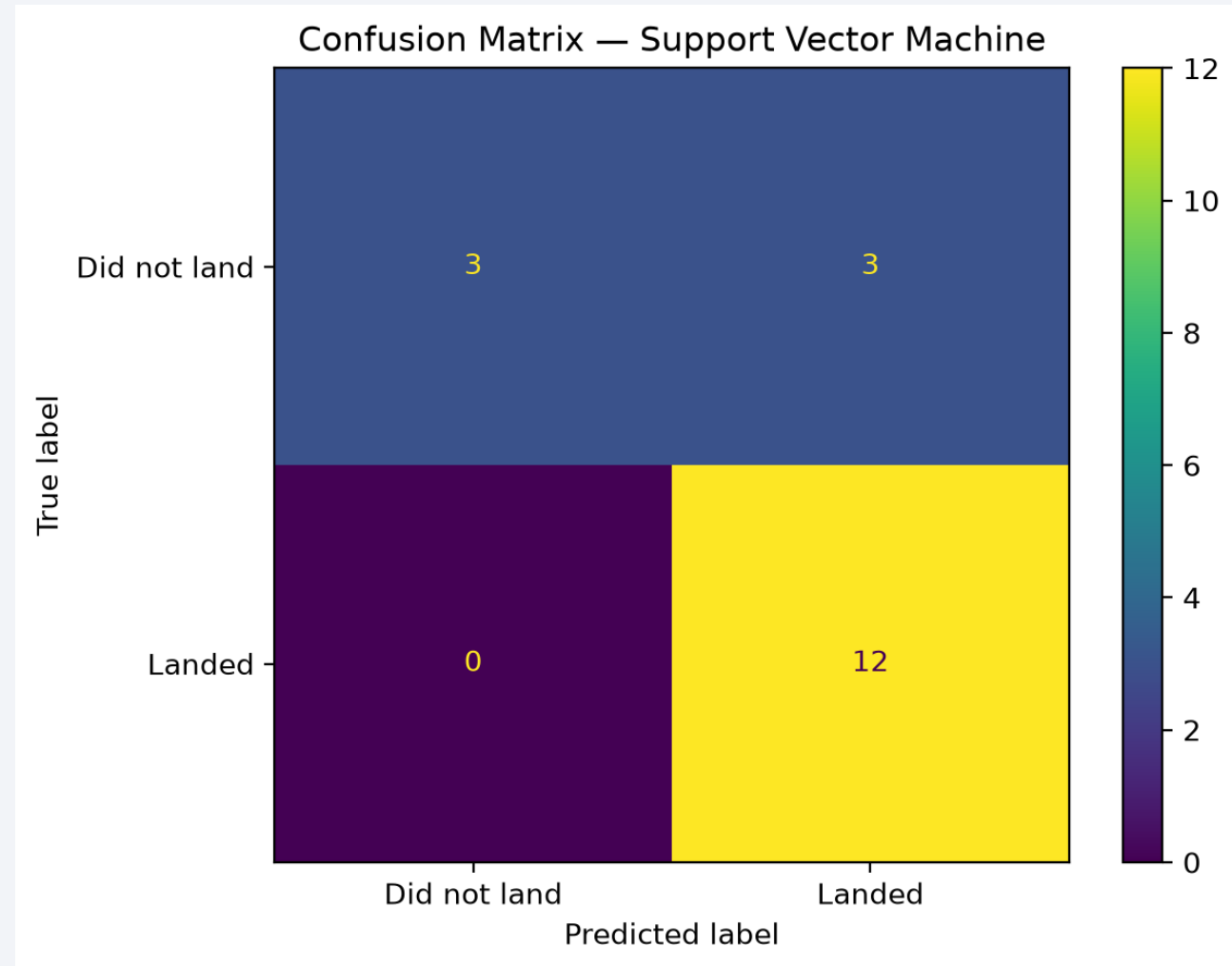
- Logistic Regression, SVM, and KNN each achieved **83.33% test accuracy**
- SVM had the highest CV accuracy among the tied test leaders: **84.82%**
- Decision Tree showed weaker test generalization at **72.22%**



Confusion Matrix

Key Findings

- **12 true positives** – successful landings correctly predicted
- **3 true negatives** – unsuccessful landings correctly predicted
- **3 false positives** – unsuccessful landings predicted as successful
- **0 false negatives** – no successful landings were missed
- **Test accuracy: 83.33%**



Conclusions

- Falcon 9 landing success generally **improved with flight experience and over time**
- **Launch site, orbit, and payload mass** all showed relationships with landing performance
- Interactive maps and dashboards helped reveal **site-level and payload-dependent patterns**
- Logistic Regression, SVM, and KNN each achieved **83.33% test accuracy**
- **SVM was selected for final reporting** with **84.82% cross-validation accuracy**
- The main prediction error was **false positives**, while all successful landings in the test set were identified

Appendix – Supporting Project Assets

- **Notebooks 01–08:** complete analysis workflow
- **Datasets:** cleaned and wrangled Falcon 9 data
- **Interactive outputs:** Folium maps and Plotly Dash visualizations
- **Model outputs:** tuning results, accuracy comparisons, and confusion matrices
- **GitHub repository:** <https://github.com/Zephyr923/SpaceX-Falcon9-Capstone>

Thank you!

